BY ORDER OF THE COMMANDER
AIR FORCE MATERIEL COMMAND

DEPARTMENT OF THE AIR FORCE
INSTRUCTION 21-101

AIR FORCE MATERIEL COMMAND
Supplement

ADDENDUM D

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Maintenance

UNITED STATES SPACE FORCE (USSF) NON STANDARD ORGANIZATION (NSO) AIRCRAFT AND EQUIPMENT MAINTENANCE MANAGEMENT

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This publication implements DAFI 21-101, *Aircraft and Equipment Maintenance Management*, and AFMCSUP servicing MAJCOM guidance for United States Space Force (USSF) Aircraft and Equipment, and Munitions Maintenance. This publication applies to USSF units designated as Non Standard Organizations (NSOs) in **paragraph 1.2** and those with USSF contractual obligation, when associated with AFMC where AFMC is the servicing MAJCOM. Note: All contractor requirements contained within this Instruction must be contained within the contract/grant/agreement to be enforceable. This publication applies to Air National Guard Bureau and Air Force Reserve Command units if assigned or associated with USSF where USSF is the lead. **Note:** This Addendum scopes the NSOs requirements in meeting the intent of the DAFI based on their mission. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with IAW AFI 33-322, *Records Management and Information Governance Program*, and disposed of IAW the Air Force Records Information

Management System Records Disposition Schedule located in the Air Force Records Information Management System . This publication may be supplemented at any level, but all direct supplements must be routed to the Office of Primary Responsibility (OPR) of this publication for coordination prior to certification and approval at **AFMC.A4M.Workflow@us.af.mil** and are published in the e-Publishing website. The authorities to waive wing/unit level requirements in this publication are identified with a Tier "T-2" number following the compliance statement. See DAFI 90-160, *Publications and Forms Management,* and DAFMAN 90-161*, Publishing Processes and Procedures*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the Publication OPR for non-tiered compliance items. Submit recommendations for change or improvement to this instruction on DAF Form 847, *Recommendation for Change of Publication*, IAW DAFI 90-160 and DAFMAN 90-161. For questions on interpreting this instruction, first contact your Quality Assurance (QA) Office/activity or Contracting Officer's Representative (COR). Send questions to the Office of OPR at AFMC/A4M Workflow at **AFMC.A4M.Workflow@us.af.mil**, 4375 Chidlaw Rd, Bldg 262 Room C114 WPAFB OH 45433-5006. Organizations that implement processes that were not identified as applicable in this Addendum or gain those functions will follow the guidance in DAFI 21-101 or submit a deviation in the form of a waiver request. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

1.1. (USSF) Introduction. The NSO applicability to DAFI 21-101 and AFI 21-101 AFMC Supplement is subjective; therefore, this instruction prescribes the basic programs to be used in NSOs (i.e., list of requirements) and provides senior management, middle management, maintenance supervisors, and technicians direction for accomplishment of aircraft maintenance management. The affected NSOs have distinct mission areas requiring development of separate guidance. The NSOs are expected to effectively use the resources assigned to ensure mission accomplishment. NSOs are required to establish guidance and may use additional management procedures not specifically prohibited by this instruction, any technical order, or other applicable instruction. Supervisors at all levels are responsible to ensure this directive is fully implemented. In the event of a conflict between this instruction and Technical Orders (TOs), TOs will take precedence.

1.2. (USSF) Organization. Assigned missions within USSF are not organized under the standard Aircraft Wing Organization concept IAW AFI 38-101,. USSF does not have assigned aircraft; transient aircraft and support equipment maintenance is a contracted function. Exception: TA at Space Base Delta 2 (Buckley SFB) is an ANG operation. Therefore, the USSF organizations assigned responsibilities and oversight for aircraft/munitions maintenance management are designed as Non-Standard Organizations (NSOs) requiring deviations (applicability, variance, and/or differences in organizational placement of responsibilities/processes) from DAFI 21-101 and the AFMC supplement as the servicing MAJCOM. DEV – Deviation to provide/scope applicability for NSO aircraft maintenance in an addendum(s) to the MAJCOM supplement is approved by USSF/AFMC as applicable. The following organizations are designated as NSOs: Space Base Delta 1, Peterson SFB; Pituffik Space Base; Space Base Delta 2, Buckley SFB; Space Launch Delta 45, Patrick SFB; Space Launch Delta 30, Vandenberg SFB; and Rocket System Launch Program, Camp Navajo. Applicability for these NSOs and those with USSF contractual

obligation, when associated with AFMC where AFMC is the servicing MAJCOM, are in **Table 1.3** USSF Applicability. **(T-2)**

1.6. (USSF) Maintenance Discipline. Comply with all written guidance and TOs.

1.9. (USSF) Publications. At a minimum, the affected NSOs will: Publish specific policy guidance (e.g., Supplement/Addendum/Operating Instruction(s) (OIs) in the appropriate publication series as required for the programs outlined in this instruction **Table 1.3**. **(T-2)**

1.10. (USSF) Maintenance Training. Additional training required when assigned as a Contracting Officer's Representative (COR), see Department of Defense Instruction (DoDI) 5000.72, *DoD Standard for Contracting Officer's (COR) Certification*, Enclosure 3, **para 1**(c), and AFI 63-138, *Acquisition of Services*, for training requirement, duties, and responsibilities.

1.12. (USSF) Maintenance Information Systems (MIS). NSOs will: Use MIS or Information Systems (IS) as directed by TOs/AFI or submit a deviation or a waiver. **(T-2)**

1.13. (USSF) General Safety Guidance. Safety and Security of Explosives, Electrostatic Discharge (ESD), and Lightning Protection.

1.13.3. **(Added)** NSOs working with, handling or storing explosive items must comply with AFI 31-101, *Integrated Defense (ID)*; DoD 5100.76-M, *Physical Security of Sensitive Conventional Arms, Ammunition, And Explosives* (AA&E); Defense Explosives Safety Regulation (DSER) 6055.09_AFMAN 91-201, *Explosives Safety Standards*; AFMAN 21-200, *Munitions and Missile Maintenance Management*; DAFMAN 21-201, *Munitions Management*, and applicable host unit instructions.

1.13.4. **(Added)** Rocket System Launch Program (RSLP) at Camp Navajo Specific; Safety and Security of Explosives-safety covered in each motor specific Standard Operating Procedures , e.g., Arizona Department of Emergency and Military Affairs Ordnance Operations Department (AZDEMA/OOD) Minuteman 1st Stage Operations paragraph one covers grounding and other asset specific safety requirements. Army Regulation (AR) 190-11, *Physical Security of Arms Ammunition and Explosives*, is guidance covering Security of explosives. All work area with explosive operations vetted by QA per local Standard Operating Procedures, Department of the Army Pamphlet (DA PAM) 742-1, *Ammunition Surveillance Procedures* and DA PAM 385-64, *Ammunition and Explosive Safety Standards*.

1.13.5. **(Added)** Electrostatic Discharge (ESD) Program. NSOs working on items that meet ESD criteria as defined in TO 00-25-234, *General Shop Practice Requirements for the Repair, Maintenance, and Test of Electrical Equipment*, must comply with TO 00-25-234 requirements.

1.13.6. **(Added)** RSLP at Camp Navajo Specific; ESD-QA has oversight. Lightning Protection System (LPS) testing, ESD covered in Navajo (NA)-0000-A-801, *Safety Program Manual*.

1.18. (USSF) Performance-Based Activities. Contract Surveillance, see *Multi-Functional Team (MFT) Resources Guide,* hosted at AFPEO/CM SharePoint®*:* **https://usaf.dps.mil/sites/SAFAQ/projects/Services/MFT/MFT.aspx**

1.18.1.4. **(USSF)** Reference AFI 63-138 para 2.15, Quality Assurance Program Coordinator responsibilities for the coordination and management of the organization's contract quality assurance program.

Table 1.3. (Added) USSF Applicability.

Paragraph	Space Base Delta 1, Peterson SFB (FIELDCOM SpOC)	Space Launch Delta 45, Patrick SFB (FIELDCOM SSC)	Space Launch Delta 30, Vandenberg SFB (FIELDCOM SSC)	Pituffik Space Base (formerly Thule, AB) (FIELDCOM SpOC)	Space Base Delta 2, Buckley SFB (FIELDCOM SpOC)	Rocket System Launch Program, (RSLP) Camp Navajo, AZ (FIELDCOM SSC)
1.6. Maintenance Discipline.	Yes	Yes	Yes	Yes	Yes	Yes
1.9. Publications. (Development)	Yes	Yes	Yes	Yes	Yes	Yes
1.10. COR Training	Yes	Yes	Yes	Yes	ANG Operation	No (*****)
1.13. General Safety	Yes	Yes	Yes	Yes	Yes	Yes
1.18. Performance-Based Activities. (e.g., Contract Surveillance)	Yes	Yes	Yes	Yes	ANG Operation	No
Chapter 2. Roles and Responsibilities	Yes	Yes	Yes	Yes	Yes	Yes
4.9.5. TA	Yes	Yes	Yes	Yes	ANG Operation	N/A
6. Quality Assurance (QA).	No (Contracted)	No (Contracted)	No (COR for TA/AGE Contract)	No (Contracted)	ANG Operation	Yes*
QA, Conventional Munitions Program	No **	Yes** (See table 6.1)	No **/***	No **	No **	Yes** (see table 6.1)

Chapter 7, Impoundment Procedures.	Yes	Yes	Yes	Yes	Yes	Yes, ****
Chapter 8. Tool and Equipment Management. (8.2. Guidelines for Program Management.	Yes	Yes	Yes	Yes	Yes	Yes
11.8. Foreign Object Damage (FOD) Prevention Program.	Yes	Yes	Yes	Yes	Yes	Yes, non critical
11.9 Dropped Object Prevention (DOP) Program.	Yes	Yes	Yes	Yes	Yes	N/A
11.28 Crash Damaged or Disabled Aircraft Recovery (CDDAR) Program	Yes	Yes	Yes	Yes	Yes	N/A

(*) Note: RSLP (SSC/AAMX) at Camp Navajo utilizes Table 6.1 small and unique guidance and use of the Quality Assurance Specialist, Ammunition Surveillance Program

(**) Note: Units with Command Code 1M & 1S fall under responsibility for HQ AFMC Munitions oversight. Any other Command codes fall under the applicable host base munitions support.

(***) Note: 576th Flight Test Sq, AFGSC tenant unit, does a QA program at Space Launch Delta 30 for test ICBM Missile launches.

(****) Note: Exception: RSLP at Camp Navajo will address/determine the program used in lieu of this program in local guidance.

(*****) Note: Currently performed by government and applicable when activities become contracted.

2.1.3. **(Added)** Establish written policy if responsibilities differ from a standard DAFI 21-101 structure. Indicate the responsibilities for key leaders involved in aircraft and equipment maintenance management activities. Due to diversity of organizational structures, responsibilities are assigned at the appropriate level as applicable. (T-2)

2.4.49.1. **(Added)** When issuing or renewing a TA contract, owning/requiring activity will use the Standardized TA Services Performance Work Statement. Additionally, when issuing, renewing, or changing a TA contract the updated Standardized TA Services Performance Work Statement will be sent to AFMC/A4M Workflow at **AFMC.A4M.Workflow@us.af.mil** for review and coordination. Link to the TA Sharepoint® page: **https://usaf.dps.mil/sites/aficc/afcc/AFICC/763ESS/TAS/SitePages/Home.aspx** and for assistance email the TA Organizational box at **763SCONS.PKB.AFTransientAlert@us.af.mil**. **(T-2)**

4.9. (USSF) Maintenance Flight. Exception: No maintenance flight assigned.

4.9.5. **(USSF)** TA is a contracted function within USSF, Exception: TA at Space Base Delta 2 (Buckley SFB) is an ANG operation. USSF NSOs with a required TA will ensure the contract contains responsibilities for recovers, services, inspects, maintains, and launch of transient aircraft. (T-2). Contracted TA processes/procedures may vary based on contract/Performance Work Statement, see Standardized TA Services Performance Work Statement requirements in **paragraph 2.4.49.1** of this Addendum.

6.1. General. (USSF) (DEV) NSOs will establish QA office function and policy. **(T-2)**At a minimum, the policy must address the roles and responsibilities of key personnel involved in the QA Program to include, as applicable, the Commander, Vice Commander, Logistics Manager, site QA Superintendent, and site QA inspector's roles. **(T-2)**

6.6.1.1. **(Added)** RSLP at Camp Navajo Specific; QA functions are managed by Chief of surveillance. Munitions Inspectors follow Quality Assurance Specialist, Ammunition Surveillance Quality Assurance Specialist, Ammunition Surveillance training surveillance plan for certification. Unit will address specifics in a local OI. **(T-2)**

6.7.2.3.1. **(USSF)** NSOs will ensure aircraft and equipment, including munitions, are inspected IAW TOs and directives. **(T-2)**

6.7.2.3.1.1. **(Added-DEV)** Units with Munitions related workload will put in writing procedures/processes to support **Table 6.1** Applicability Matrix if different than already documented DoD/AF/AFMC policy. **(T-2)**

Table 6.1. (Added) USSF Conventional Munitions QA Program Applicability Matrix.

AFMC Conventional Munitions QA Program Applicability Matrix	Rocket System Launch Program (RSLP), Camp Navajo, AZ	Space Launch Delta 45 , Patrick SFB
10. Accountability	X	X

11. Storage practices, security, and safety	X	X
12. Inspection	X	X
13. Materiel and handling and test equipment	X	X
14. Stockpile management	X	X
15. Training programs	X	X
16. Infrastructure (LPS, grounds, and bonds)	X	X
17. CTKs, tools, and support equipment	X	X
Note: If "x" you will follow prescribed guidance.		

7.2.1.1.1. **(Added)** USSF NSOs will ensure development of a local instruction for local impoundment procedures. **(T-2)** Exception: RSLP at Camp Navajo will address/determine the program used in lieu of this program in local guidance.

8.1. (USSF) Tool and Equipment Management. NSOs will ensure a coordinated wing or base instruction is developed that implements tools and equipment management procedures. **(T-2)**

8.1.1. **(Added)** RSLP at Camp Navajo will define local tool program in a local instruction.

11.8. (USSF) Foreign Object Damage (FOD) Prevention Program. NSOs will manage the Foreign Object Damage (FOD) Programs. **(T-2)**

11.8.2.1. **(Added)** There are two categories of Foreign Object (FO) areas: Critical and non-critical. Foreign Object critical areas are areas where aerospace maintenance, testing, and operations are performed (e.g., jet engine maintenance, fuel cell maintenance, major sub-assembly maintenance, and support equipment maintenance). Non-critical Foreign Object areas are all other areas not defined previously. NSOs shall identify and document critical/non-critical Foreign Object areas. **(T-2)**

11.8.2.1.1. **(Added)** RSLP at Camp Navajo will follow Non-critical FOD program and will define specifics in local instruction.

11.8.3.11. **(USSF)** Courtesy copy the servicing MAJCOM FOD monitor at AFMC/A4M Workflow at **AFMC.A4M.Workflow@us.af.mil** or HQ AFMC/A4M Propulsion Management SharePoint® site: **https://usaf.dps.mil/teams/22355/Lists/ALC%20FOD%20Incident%20Report/Standard%20View.aspx** . **(T-2)**

11.8.6.4.4. **(Added)** Damage beyond blend repair, where hardware may be suspect for cause, and/or further unserviceable damage exists beyond the 1st stage fan will be considered reportable FOD. FOD found on the 1st stage fan which can be blended IAW with applicable technical data may be considered non-reportable. **(T-2)**

11.8.6.6. **(USSF)** HQ AFMC/A4M Propulsion Functional Manager shall assist in resolving questionable FOD issues. Send all correspondence to AFMC/A4M Workflow at **AFMC.A4M.Workflow@us.af.mil**.

11.8.6.7. **(USSF)** Send all correspondence to AFMC/A4M Workflow at **AFMC.A4M.Workflow@us.af.mil**. **(T-2)**

11.8.6.7.2.1. **(Added)** MDS specific flying hours source data is defined as the flying hours flown as identified in the Maintenance Information System (MIS). **(T-2)**

11.8.6.7.2.2. **(Added)** If the FOD incident is deemed a mishap IAW AFI 91-204, the FOD monitor will work in tandem with the Wing SE office to properly report the mishap. **(T-2)**

11.8.7. **(Added)** The FOD meeting may be combined with other meetings. **(T-2)**

11.9. (USSF) Dropped Object Prevention (DOP) Program. NSOs will manage the Dropped Object Prevention (DOP) Programs. **(T-2)**

11.9.3.1.1. **(USSF)** Courtesy copy the servicing MAJCOM DOP monitor at AFMC/A4M Workflow at **AFMC.A4M.Workflow@us.af.mil**. **(T-2)**

11.28. (USSF) Crash Damaged or Disabled Aircraft Recovery (CDDAR) Program. USSF Installation/Space Delta/Commanders with active airfields/runways, and flying missions are responsible for sustaining a CDDAR capability.

11.28.2.1.2. **(USSF)** HQ AFMC/A4MM is the MAJCOM CDDAR OPR. Send all correspondence to AFMC/A4M Workflow at **AFMC.A4M.Workflow@us.af.mil**.

C. M. VON HOFFMAN, Major General, USAF
Director of Logistics, Civil Engineering, Force Protection, and Nuclear Integration

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING DOCUMENTS

References

AFI 33-322, *Records Management and Information Governance Program,* 28 July 2021

AFI 38-101, *Manpower and Organization*, 29 Aug 2019

AFI 63-138, *Acquisition of Services*, 30 September 2019

AFMAN 21-200, *Munitions and Missile Maintenance Management*, 9 Aug 2018

AR 190-11, *Physical Security of Arms Ammunition and Explosives*, 17 Jan 2019

DAFI 21-101, *Aircraft and Equipment Maintenance Management*, 16 Jan 2020

DAFI 31-101, *Integrated Defense (ID)*, 25 Mar 2020

DAFI 90-160, *Publications and Forms Management*, 14 April 2022

DAFI 91-204, *Safety Investigations and Reports*, 10 March 2021

DAFMAN 21-201, *Munitions Management*, 03 May 2022

DAFMAN 90-161*, Publishing Processes and Procedures*, 15 April 2022

DA PAM 385-64, *Ammunition and Explosive Safety Standards*. 10 Oct 2013

DA PAM 742-1, *Ammunition Surveillance Procedures*, 22 Nov 2016

DESR 6055.09_AFMAN91-201, *Explosives Safety Standards*, 28 May 2020

DoD 5100.76-M, *Physical Security of Sensitive Conventional Arms, Ammunition, And Explosives (AA&E)*; 17 April 2012

DoDI 5000.72, *DoD Standard for Contracting Officer's Representative (COR) Certification,* 26 March 2015 incorporating Change 2, November 6, 2020

NA-0000-A-801, *Safety Program Manual,* 01 Jun 2018

TO 00-25-234, *General Shop Practice Requirements for the Repair, Maintenance, and Test of Electrical Equipment*, 27 October 2021

Prescribed Forms

None

Adopted Forms

None

Abbreviations and Acronyms

ANG—Air National Guard Bureau

COR—Contracting Officer's Representative

ESD—Electrostatic Discharge

FIELDCOM—Field Commands

FOD—Foreign Object Damage

NSO—Non Standard Organization

OPR—Office of Primary Responsibility

QA—Quality Assurance

RSLP—Rocket System Launch Program

SFB—Space Force Base

SpOC—Space Operations Command

SSC—Space Systems Command

TA—Transient Alert

USSF—United States Space Force

Office Symbols

AF/A4L—Director of Logistics DCS/Logistics, Engineering, & Force Protection

AFMC/A4M—Maintenance Division

AFMC/A4MM—Maintenance Management Branch

AZDEMA/OOD—Arizona Department of Emergency and Military Affairs Ordnance Operations Department

HQ/SF COO—Space Force Chief Operations Office

Terms

Delta—(Also called "Space Base Delta or Space Launch Delta")—A component of the USSF subordinate to a field command, similar to an Air Force Wing. Responsible for Space Mission Force operations and oversight.

Field—Organizational entity below the HAF, MAJCOM or FLDCOM level. Examples include: FOA, DRU, NAF, wing or delta/garrison, etc.

FIELDCOM—Space Force Field Commands (FIELDCOM) are a major subdivision of the Space Force that are assigned a major part of the Space Force mission. Similar to an Air Force Major Command (MAJCOM), a FIELDCOM is directly subordinate to the Chief of Space Operations. FIELDCOM headquarters have tailored staffs necessary to accomplish their specific missions and provide operational leadership and supervision to assigned subordinate units, such as Mission and Space Base Deltas and squadrons. There are three Space Force Field Commands: 1) Space Operations Command (SpOC), 2) Space Systems Command (SSC), and 3) Space Training and Readiness Command (STARCOM).

Garrison—A body of guardians (troops) stationed in a particular location.

Implementing Publication—A publication that creates processes and sets standards to accomplish a goal, often by adding organization-specific material to a publication issued by a higher headquarters.

Non Standard Orgainizations (NSOs)—NSOs are organizations where their logistics/maintenance support missions are not organized under the standard AFI 21-101 Wing Organization concept in accordance with (IAW) AFI 38-101, and require supplemental guidance.

Servicing MAJCOM—Servicing MAJCOM, although a non-doctrinal term, is intended to describe the roles and responsibilities traditionally performed for Airmen by a MAJCOM. These functions typically include, but are not limited to, providing policy guidance, interpretation and, where applicable, waiver authority; professional development opportunities and guidance; developmental team representation; and functional-specific roles.

Unit—A military organization constituted by directives issued by DAF, USAF or USSF.

United States Space Force—The space service branch of the U. S. Armed Forces. One of eight U.S. uniformed services, and the world's first and currently only independent space force.